

Newington College



HSC Trial Examination, 1999

12 MATHEMATICS 4 UNIT

HSC Trial Examination, 1999

Time allowed - Three hours

(plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES:

- All questions may be attempted.
- All questions are of equal value.
- In every question, show all necessary working.
- Marks may not be awarded for careless or badly arranged work.
- A table of standard integrals is provided for your convenience. Approved silent calculators may be used.
- The answers to the eight questions in this paper are to be returned in separate bundles clearly marked Question 1, Question 2 etc.
- Each bundle must show your Candidate's number.
- The questions are not necessarily arranged in order of difficulty. Candidates are advised to read the whole paper carefully at the start of the examination.

QUESTION 1 (Start a new page.) (15 Marks)

(a) Evaluate (i) $\int_0^4 \frac{1}{\sqrt{x^2+9}} dx$ (5 marks)

(ii) $\int_2^4 \frac{dx}{x^2-2x+4}$

(b) Find (i) $\int \frac{(\sqrt{x}-1)^6}{\sqrt{x}} dx$ (5 marks)

(ii) $\int e^{-x} \cos \frac{x}{2} dx$

(c) (i) If $I_n = \int_0^{\pi/4} \tan^{2n} x dx$ prove that (5 marks)

$$I_n + I_{n-1} = \frac{1}{2n-1}, \text{ for } n \geq 1.$$

(ii) Hence evaluate $\int_0^{\pi/4} \tan^6 x dx$.

QUESTION 2 (Start a new page.) (15 Marks)

(a) (i) If $Z = -1 + \sqrt{3}i$, find $|Z|$ and $\arg Z$. (4 marks)

(ii) Hence evaluate $(-1 + \sqrt{3}i)^9$.

(b) (i) Express the value of $(-1 + \sqrt{3}i)(1+i)$ in the form $a+ib$. (4 marks)

(ii) Hence, or otherwise, find the exact value of $\cos \frac{11\pi}{12}$.

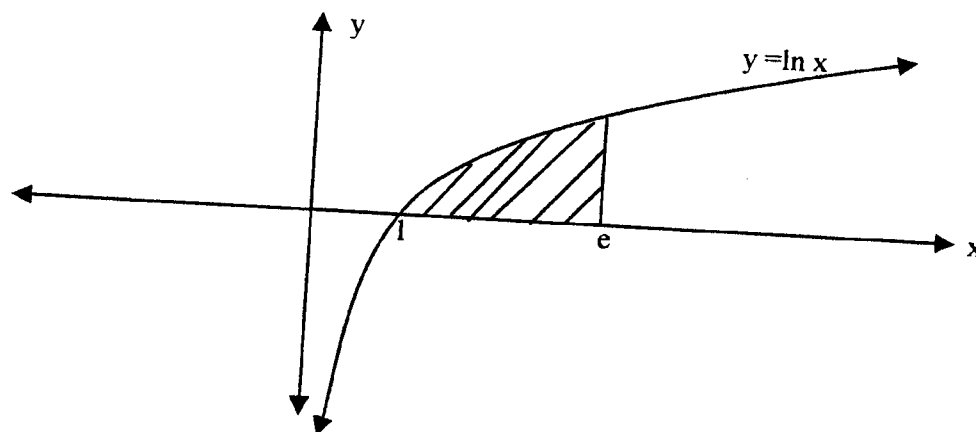
(c) Graph the region in the complex plane for which (3marks)

$$2 < |z-1+2i| < 3.$$

(d) If $|z| < \frac{1}{2}$, show that $|(1+i)z^3 + iz| < \frac{3}{4}$. (4 marks)

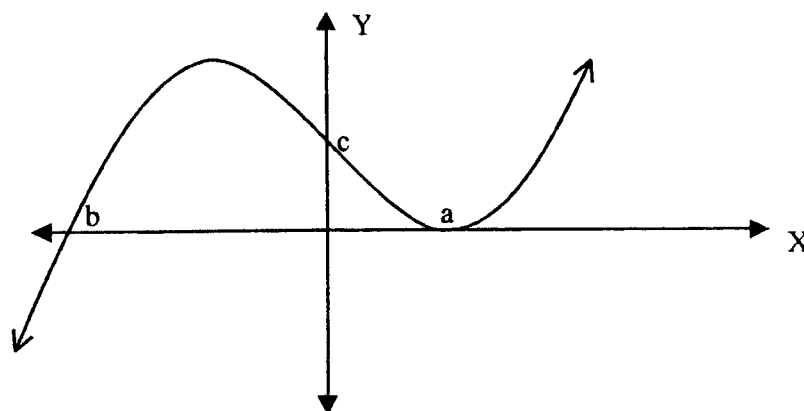
QUESTION 3 (Start a new page.) (15 Marks)

- (a) Consider the polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$ (7 marks)
- (i) If $P(x)$ has the zeros $a + bi$, $a - 2bi$, where a and b are real, find the values of a and b .
- (ii) Hence, find all the zeros of $P(x)$ over the complex field and express $P(x)$ as the product of two factors.
- (b) (i) If α is a double root of $f(x) = 0$, show that α is a root of $f'(x) = 0$. (4 marks)
- (ii) Show that if the equation $x^n + px + q = 0$ has a double root α (where α , p , q are real non-zero constants, and n is an integer with $n \geq 2$), then:
- $$\alpha = \frac{qn}{p(1-n)}$$
- (c) Using the method of cylindrical shells, find the volume generated by revolving the region bounded by $y = \ln x$, the x -axis and $1 \leq x \leq e$, about the y -axis. (4 marks)



QUESTION 4 (Start a new page.) (15 Marks)

(a)



The graph of the function $y = f(x)$ is sketched above. On separate number planes sketch the graphs of:

(10 marks)

(i) $y = f(-x)$

(ii) $y^2 = f(x)$

(iii) $y = f(|x|)$

(iv) $y = \frac{1}{1-f(x)}$

(b) (i) Resolve $\frac{1}{(x+1)(x^2+4)}$ into partial fractions.

(5 marks)

(ii) Use this result to show that

$$\int_0^2 \frac{1}{(x+1)(x^2+4)} dx = \frac{1}{10} \left(\frac{\pi}{4} + \log \frac{9}{2} \right).$$

QUESTION 5 (Start a new page.) (15 Marks)

- (a) (i) Consider the rectangular hyperbola $xy = c^2$, where $c > 0$. Prove that (8 marks)
the equation of the chord joining the points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$,
where $0 < p < q$, is given by: $x + pqy = c(p + q)$.

- (ii) The chord PQ intersects the x and y axes in A and B respectively.
Prove that $AP=BQ$.

- (iii) Show that the area enclosed by the hyperbola $xy = c^2$ and the chord PQ is:

$$\frac{c^2(q^2 - p^2)}{2pq} + c^2 \ln \frac{p}{q} \text{ square units.}$$

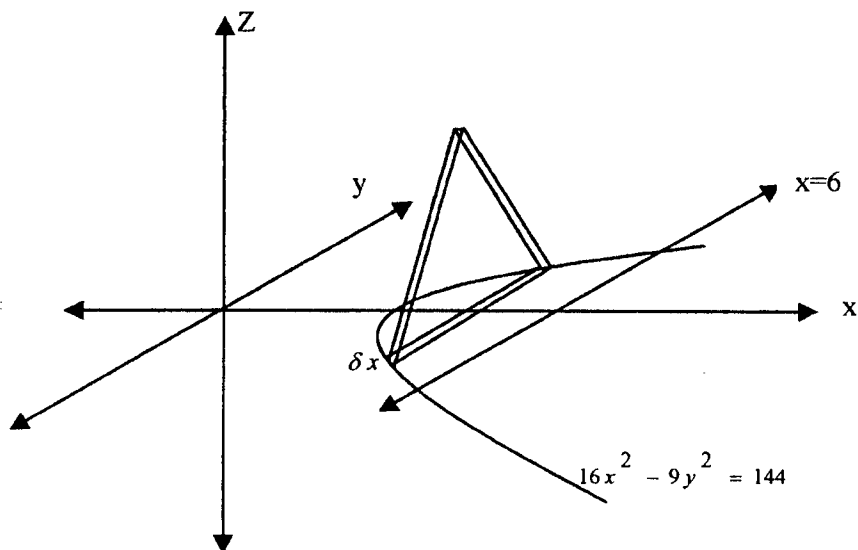
- (b) A solid has as its base the area bounded by the hyperbola $16x^2 - 9y^2 = 144$ and the line $x=6$.
Every cross-section of this solid perpendicular to the x-axis is an isosceles triangle of
altitude 3.

- (i) Show that the volume V of the resulting solid is given by: (7marks)

$$V = 4 \int_3^6 \sqrt{x^2 - 9} \, dx$$

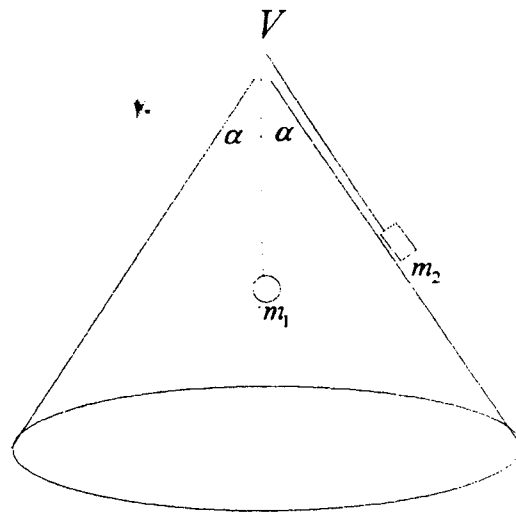
- (ii) Hence, show that:

$$V = 36\sqrt{3} - 18\log(2 + \sqrt{3}).$$



QUESTION 6 (Start a new page.) (15 Marks)

- (a) A hollow cone whose vertical angle is 2α is fixed with its axis vertical and with vertex V uppermost. A light inextensible string passes without friction through a small hole at V and carries a particle P_1 of mass m_1 kg at one end so that P_1 hangs vertically at rest inside the cone. The other end of the string carries a particle P_2 of mass m_2 kg, which moves in a horizontal circle at constant angular velocity ω on the smooth outer surface of the cone, at a vertical depth h metres below V .



(7 marks)

- (i) Prove $m_2(h\omega^2 \sin^2 \alpha + g \cos^2 \alpha) = m_1 g \cos \alpha$.
- (ii) Find the magnitude of the force exerted by the surface of the cone on P_2 , and hence deduce that $h\omega^2 < g$.
- (b) Two particles move in the same vertical line in a medium whose resistance, per unit mass, varies as the velocity. One particle is projected vertically upwards from the ground with initial velocity u , and starting at the same instant, the other particle falls from a height, h metres.

(8 marks)

- (i) For the particle which is projected vertically upwards from the ground, show that the expression for its height (x) metres after a time (t) seconds is given by:

$$x = \frac{g + ku}{k^2} (1 - e^{-kt}) - \frac{gt}{k},$$

where g is the acceleration due to gravity and k is a constant.

- (ii) Assuming that the height of the falling particle is given by

$$h - \frac{gt}{k} - \frac{ge^{-kt}}{k^2} + \frac{g}{k^2},$$

prove that the particles meet after a time (T), where:

$$T = \frac{1}{k} \log \left(\frac{u}{u - kh} \right).$$

QUESTION 7 (Start a new page.) (15 Marks)

- (a) A group of men and women is seated randomly around a circular table. What is the probability that none of the men are sitting next to each other if there are:
- (i) 3 men and 4 women? (4 marks)
- (ii) n men and $n+1$ women?
- (b) If a, b, c and d are positive real numbers, prove that: (6 marks)
- (i) $\frac{a+b}{2} \geq \sqrt{ab}$,
- (ii) $(a+b+c+d)^2 \geq 4(ac+bc+bd+ad)$,
- (iii) $(a+b+c+d)^2 \geq \frac{8}{3}(ab+ad+bc+cd+bd+ac)$.
- (c) A sequence is defined by the relationship $a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$ where $a_1 = 1$ and n is a positive integer. (5 marks)
- (i) Show, using mathematical induction, that $\frac{a_n - \sqrt{2}}{a_n + \sqrt{2}} = \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}}\right)^{2^{n-1}}$.
- (ii) Hence find the limiting value of a_n as n becomes large.

QUESTION 8 (Start a new page.) (15 Marks)

- (a) Find all real
- x
- such that

(3 marks)

$$3\sqrt{x(1-x)} < |x-2|$$

- (b) If a curve is given by
- $y = f(x)$
- , where
- $f(x)$
- has a continuous derivative in the open interval between
- $x = a$
- and
- $x = b$
- then the length is given by

$$\int_a^b \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{1}{2}} dx \quad (4 \text{ marks})$$

Use this result to prove that the circumference of a circle, with radius r , is equal to $2\pi r$.

- (c) (i) Prove that for
- $t \neq -1$
- ,

$$1 - t + t^2 - t^3 + \dots + t^{2n} = \frac{1}{1+t} + \frac{t^{2n+1}}{1+t}.$$

(8 marks)

- (ii) Hence deduce that for
- $x > -1$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + \frac{x^{2n+1}}{2n+1} - \int_0^x \frac{t^{2n+1}}{1+t} dt.$$

- (iii) For
- $0 \leq x \leq 1$
- , find

$$\lim_{n \rightarrow \infty} \int_0^x \frac{t^{2n+1}}{1+t} dt, \text{ giving reasons for your answer.}$$

- (iv) Hence find an infinite series converging to
- $\ln 2$
- .

END OF PAPER

4. Unit

Q1 a) i) $\int_0^4 \frac{1}{\sqrt{x^2+9}} dx = \left[\ln \left| x + \sqrt{x^2+9} \right| \right]_0^4$

$$= \ln[4+5] - \ln 3$$

$$= \underline{\ln 3}$$

ii) $\int_2^4 \frac{dx}{x^2-2x+4}$

$$= \int_2^4 \frac{dx}{(x-1)^2+3} = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x-1}{\sqrt{3}} \right) \Big|_2^4$$

$$= \frac{1}{\sqrt{3}} \tan^{-1} \sqrt{3} - \frac{1}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}} \left[\frac{\pi}{3} - \frac{\pi}{6} \right] = \underline{\underline{\frac{\pi}{6\sqrt{3}}}}$$

b) (i) $\int \frac{(\sqrt{x}-1)^6}{\sqrt{x}} dx$ let $u = \sqrt{x} = x^{\frac{1}{2}}$

$$du = \frac{1}{2} x^{-\frac{1}{2}} dx = \frac{dx}{2\sqrt{x}}$$

$$= \int (u-1)^6 \cdot 2 du = 2 \frac{(u-1)^7}{7} + C$$

$$= \underline{\underline{\frac{2(\sqrt{x}-1)^7}{7} + C}}$$

ii) $\int e^{-x} \frac{\cos x}{2} dx = \frac{e^{-x}}{2} \sin x - \int \frac{2 \sin x}{2} \cdot -e^{-x} dx$

$$= \frac{2e^{-x} \sin x}{2} + 2 \int e^{-x} \cdot \frac{2 \cos x}{2} - \int \frac{-2 \cos x \cdot e^{-x}}{2}$$

$$= \frac{2e^{-x} \sin x}{2} - \frac{4e^{-x} \cos x}{2} - 4 \int \frac{e^{-x} \cos x}{2} dx$$

$$Q1 c) (i) I_n = \int_0^{\pi/4} \tan^{2n} x \, dx = \int_0^{\pi/4} \tan^{2n-2} x \cdot \tan^2 x \, dx$$

$$I_n = \int_0^{\pi/4} \tan^{2n-2} x \sec^2 x \, dx - I_{n-1}$$

$$I_n = \frac{\tan^{2n-1} x}{2n-1} \Big|_0^{\pi/4} - I_{n-1} = \frac{1}{2n-1} - I_{n-1}$$

$$(ii) I_3 = \int_0^{\pi/4} \tan^6 x \, dx$$

$$= \frac{1}{5} - I_2$$

$$= \frac{1}{5} - \left[\frac{1}{3} - I_1 \right]$$

$$= \frac{1}{5} - \frac{1}{3} + \frac{1}{1} - I_0$$

$$\text{Now } I_0 = \int_0^{\pi/4} dx = \frac{\pi}{4}$$

$$\therefore I_3 = \frac{13}{15} - \frac{\pi}{4}$$

2 a) (i) $z = -1 + \sqrt{3}i$

$$|z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{4} = \underline{\underline{2}}$$

$$\arg z = \tan^{-1}\left(\frac{\sqrt{3}}{-1}\right) \quad 2^{\text{nd}} \text{ quad.}$$

$$= \underline{\underline{\frac{2\pi}{3}}}$$

$$(ii) \quad (-1 + i\sqrt{3})^9 = \left(2 \cos \frac{2\pi}{3}\right)^9 = 2^9 \cos 6\pi$$

$$= \underline{\underline{512}}$$

b) (i) $(-1 + \sqrt{3}i)(1 + i) = -1 - \sqrt{3} + i\sqrt{3} - i$
 $= \underline{\underline{-(\sqrt{3}+1) + i(\sqrt{3}-1)}}$ $\rightarrow$

$$(ii) \quad -1 + \sqrt{3}i = 2 \cos \frac{2\pi}{3} \quad 1+i = \sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right) = \sqrt{2} \cos \frac{\pi}{4}$$

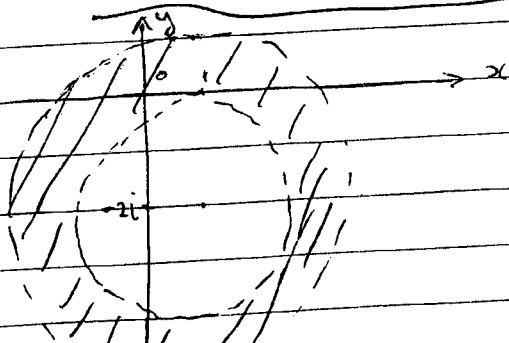
$$(-1 + \sqrt{3}i)(1 + i) = 2\sqrt{2} \cos \frac{2\pi}{3} \cos \frac{\pi}{4}$$

$$= 2\sqrt{2} \cos\left(\frac{2\pi}{3} + \frac{\pi}{4}\right)$$

$$= 2\sqrt{2} \cos \frac{11\pi}{12}$$

$$\therefore \cos \frac{11\pi}{12} = \underline{\underline{-\frac{(\sqrt{3}+1)}{2\sqrt{2}}}} \quad \text{from } \rightarrow$$

c)



Q2 d) $|z| < \frac{1}{2}$

$$|(1+i)z^3 - iz| \leq |(1+i)||z^3| + |i||z| \quad \text{from } \Delta$$

$$\leq \sqrt{2} \cdot \frac{1}{8} + \frac{1}{2}$$

Now $\frac{1}{4} > \frac{\sqrt{2}}{8}$

$$\therefore \text{LHS} \leq \frac{\sqrt{2}}{8} + \frac{1}{2} < \frac{3}{4}$$

Q3 (i) $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$

zeros $a+bi, a-2bi$ a, b real.

Since coeffs are real then $a-bi$ & $a+2bi$ are the conjugate roots

$\therefore$ roots are

$$a+ib, a-ib, a+2ib, a-2ib$$

$$a+ib+a-ib+a+2ib+a-2ib = 4$$

$$\therefore a = 1$$

$$(a+ib)(a-ib)(a+2ib)(a-2ib) = (a^2-b^2)(a^2-4b^2) = 10$$

$$\therefore (1-b^2)(1-4b^2) = 10$$

$$4b^4 - 5b^2 - 9 = 0$$

$$(4b^2 - 9)(b^2 + 1) = 0$$

$$\therefore b = \pm \frac{3}{2}$$

$$b^2 + 1 \neq 0, b \text{ real.}$$

$$a = 1, b = \frac{3}{2}$$

3 a) (ii) roots of $P(x)$ are
 $1 \pm i, 1 \pm 2i$

$$\begin{aligned} P(x) &= (x-1-i)(x-1+i)(x-1-2i)(x-1+2i) \\ &= [(x-1)^2 + 1][(x-1)^2 + 4] \\ &= [x^2 - 2x + 2][x^2 - 2x + 5] \end{aligned}$$

(i) Let $f(x) = (x-\alpha)^2 g(x)$ $g(x)$ poly

$$\begin{aligned} f'(x) &= 2(x-\alpha)g(x) + (x-\alpha)^2 g'(x) \\ &= (x-\alpha) \underbrace{[2g(x) + (x-\alpha)g'(x)]}_{\text{poly}} \end{aligned}$$

$$\therefore f'(\alpha) = 0$$

Hence α is a root of $f'(x) = 0$

i) Let $x^n + px + q = 0$ have a double root, α .
 $\therefore nx^{n-1} + p = 0$ has a root α too.

$$\begin{aligned} \text{Hence } \alpha^n + p\alpha + q &= 0 \quad \& \quad n\alpha^{n-1} + p = 0 \\ \therefore \alpha^{n-1} &= -\frac{p}{n} \end{aligned}$$

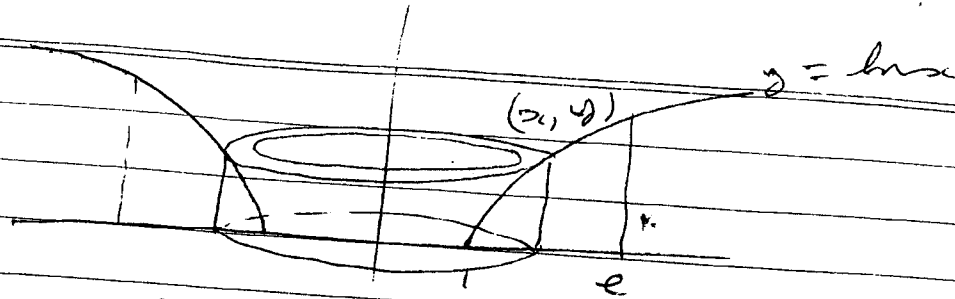
$$\alpha^n + p\alpha + q = \alpha \cdot \alpha^{n-1} + p\alpha + q = 0$$

$$= \alpha \cdot \left(-\frac{p}{n}\right) + p\alpha + q = 0$$

$$p\alpha \left[1 - \frac{1}{n}\right] + q = 0$$

$$\alpha = \frac{qn}{p(1-n)}$$

Q 3 c)



$$\delta V = \pi x^2 y - \pi (x - \delta x)^2 y$$

$$= \pi y [x^2 - (x^2 - 2x \delta x + \delta x^2)]$$

$$\approx \pi y \cdot 2x \delta x$$

δx^2 negligible $\delta x \ll x$

$$V = \sum_{n=1}^{\infty} \pi y \cdot 2x \delta x = \sum_{n=1}^{\infty} 2\pi (\ln x) x \delta x$$

$$V = \int_1^e 2\pi y x dx$$

$$= \int_1^e 2\pi x (\ln x) dx$$

$$= 2\pi \left[\ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \right]$$

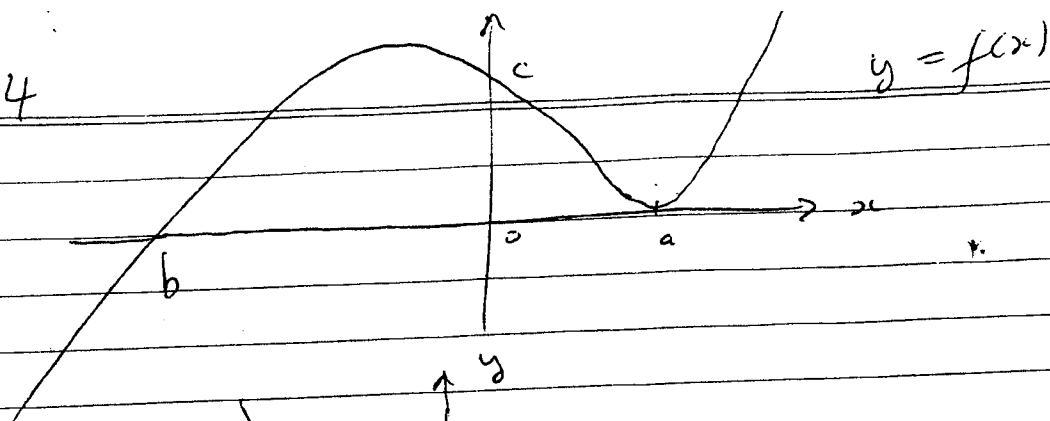
$$= 2\pi \left[\ln x \cdot \frac{x^2}{2} - \frac{x^2}{4} \right]_1^e$$

$$= 2\pi \left[\frac{e^2}{2} - \frac{e^2}{4} \right] - 2\pi \left[0 - \frac{1}{4} \right]$$

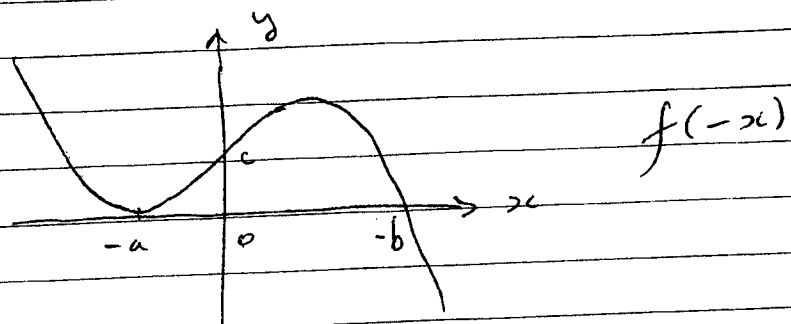
$$= 2\pi \left[\frac{e^2}{4} + \frac{1}{4} \right]$$

$$= \frac{\pi}{2} (e^2 + 1)$$

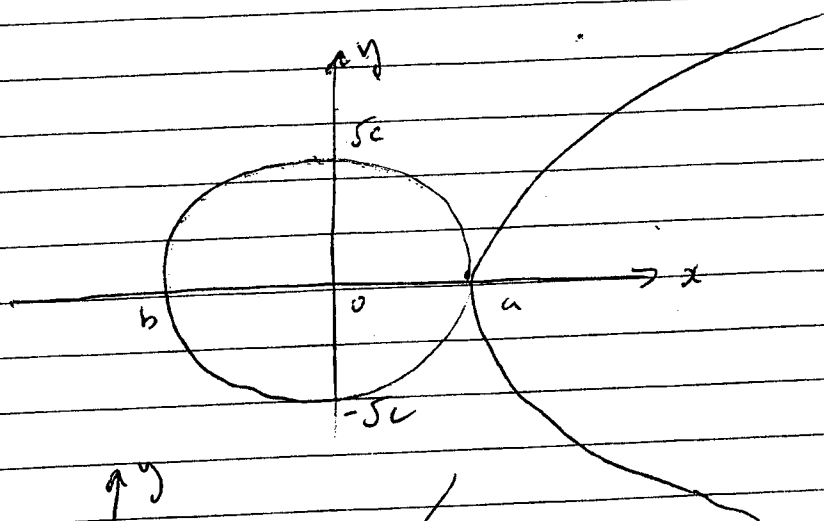
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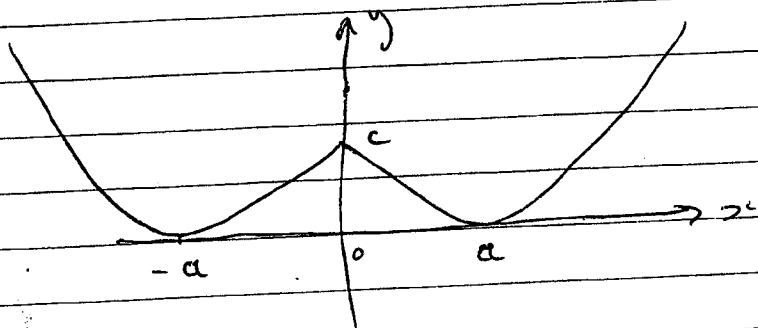
(i)



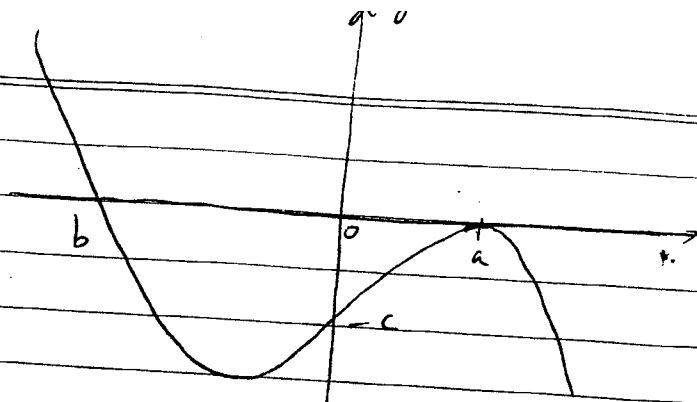
(ii) $y^2 = f(x)$



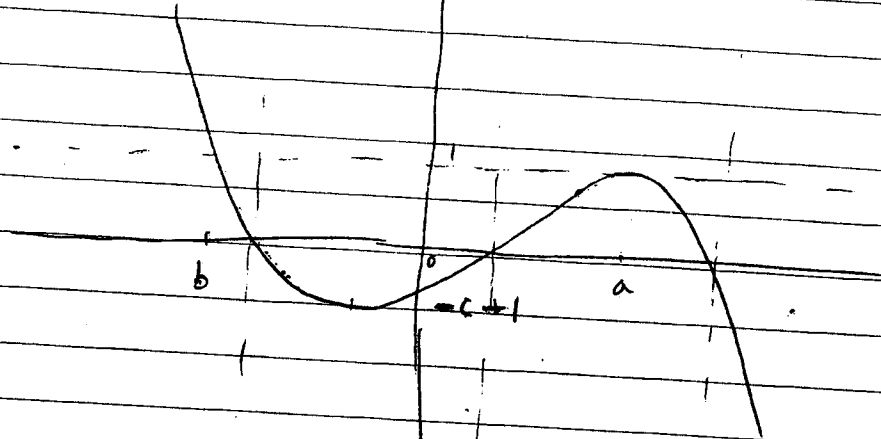
(iii)



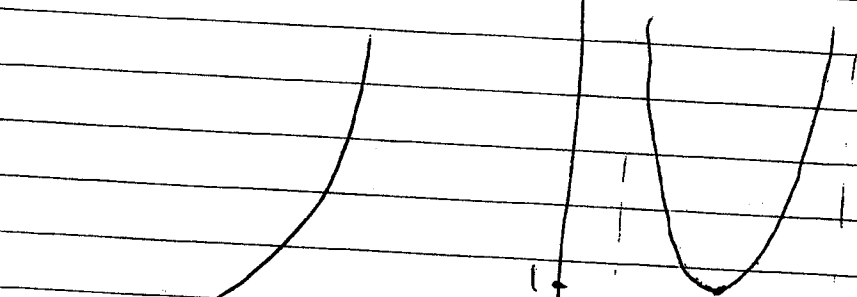
Q 4a) iv)



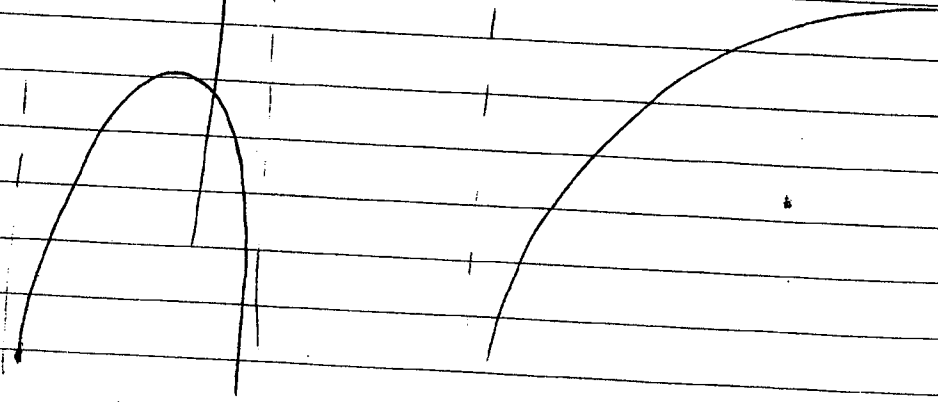
$-f(x)$



$1 - f(x)$



$\frac{1}{1 - f(x)}$



$$Q4 b) i) \frac{1}{(x+1)(x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4}$$

$$1 \equiv A(x^2+4) + (x+1)(Bx+C)$$

$$x = -1, \quad 1 = 5A \quad \therefore A = \frac{1}{5}$$

$$x = 0, \quad 1 = \frac{1}{5} \cdot 4 + C \quad \therefore C = +\frac{1}{5}$$

$$\text{Coeff } x^2 \quad 0 = A + B \quad \therefore B = -\frac{1}{5}$$

$$\frac{1}{(x+1)(x^2+4)} = \frac{1}{5(x+1)} - \frac{1}{5} \left(\frac{x-1}{x^2+4} \right)$$

$$\therefore \int \frac{1}{(x+1)(x^2+4)} dx = \frac{1}{5} \int \frac{dx}{x+1} - \frac{1}{5} \int \frac{x-1}{x^2+4} dx$$

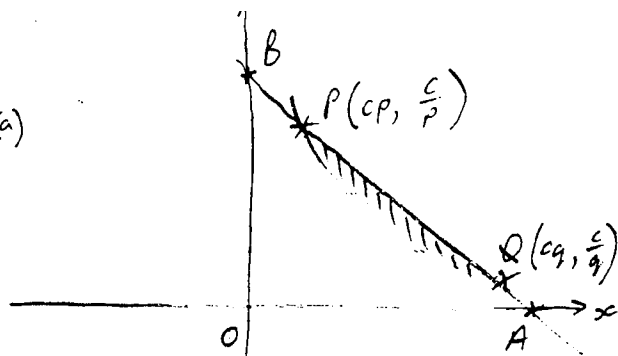
$$= \frac{1}{5} \ln(x+1) - \frac{1}{10} \ln(x^2+4) + \frac{1}{5} \cdot \frac{1}{2} \tan^{-1} \frac{x-1}{2}$$

$$= \left(\frac{1}{5} \ln 3 - \frac{1}{10} \ln 8 + \frac{1}{10} \cdot \frac{\pi}{4} \right) - \left(0 - \frac{1}{10} \ln 4 - 0 \right)$$

$$= \frac{1}{10} \left[\ln 9 - \ln 8 + \ln 4 + \frac{\pi}{4} \right]$$

$$= \frac{1}{10} \left[\ln \frac{9}{2} + \frac{\pi}{4} \right]$$

5/(a)



$$i) m_{PQ} = \frac{\frac{c}{q} - \frac{c}{p}}{cq - cp} = -\frac{1}{pq}$$

Equation PQ:

$$y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$$

$$pqy - cq = -x + cp$$

$$x + pqy = c(p+q)$$

$$ii) A(c(p+q), 0), B(0, \frac{c(p+q)}{pq})$$

$$AP = \sqrt{(c(p+q) - cp)^2 + (0 - \frac{c}{p})^2}$$

$$= \sqrt{c^2q^2 + \frac{c^2}{p^2}}$$

$$AQ = \sqrt{(0 - cq)^2 + (\frac{c(p+q)}{pq} - \frac{c}{q})^2}$$

$$= \sqrt{c^2q^2 + \frac{c^2}{p^2}}$$

$$\therefore AP = AQ$$

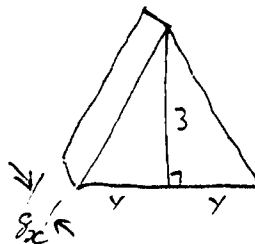
$$ii) \text{Area} = \frac{1}{2}(c(p+q) - cp)(\frac{c}{p} - \frac{c}{q}) - \int_{cp}^{cq} \frac{c^2}{x} dx$$

$$= \frac{c(q-p) \cdot c(q+p)}{2pq} - [c^2 \ln x]_{cp}^{cq}$$

$$= \frac{c^2(q^2 - p^2)}{2pq} - c^2 \ln \frac{q}{p}$$

$$= \frac{c^2(q^2 - p^2)}{2pq} + c^2 \ln(\frac{p}{q})$$

(b) Consider a typical slice



$$16x^2 - 9y^2 = 144$$

$$y = 0, x = \pm 6$$

$$y = \pm \frac{\sqrt{16x^2 - 144}}{3}$$

$$\delta V \doteq 3y \delta x$$

$$V \doteq \sum_{x=3}^{x=6} 3y \delta x$$

$$= \lim_{\delta x \rightarrow 0} \sum_{x=3}^{x=6} 3y \delta x$$

$$= \int_3^6 3y dx$$

$$= \int_3^6 \sqrt{16x^2 - 144} dx$$

$$= 4 \int_3^6 \sqrt{x^2 - 9} dx$$

$$(ii) \text{ Let } x = 3 \sec \theta$$

$$x = 3, \theta =$$

$$\therefore dx = 3 \sec \theta \tan \theta d\theta$$

$$x = 6, \theta =$$

$$\therefore V = 4 \int_0^{\frac{\pi}{3}} \sqrt{9(\sec^2 \theta - 1)} \cdot 3 \sec \theta \tan \theta d\theta$$

$$= 36 \int_0^{\frac{\pi}{3}} \tan \theta (\sec \theta \tan \theta) d\theta$$

$$\text{Now } \int_0^{\frac{\pi}{3}} \tan \theta (\sec \theta \tan \theta) d\theta$$

$$= \left[\tan \theta \cdot \sec \theta \right]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \sec \theta \cdot \sec^2 \theta d\theta$$

5/(b) cont.

$$\int_0^{\frac{\pi}{3}} \sec \theta \tan^2 \theta = (\sqrt{3} \cdot 2 - 0) - \int_0^{\frac{\pi}{3}} \sec \theta (1 + \tan^2 \theta) d\theta$$

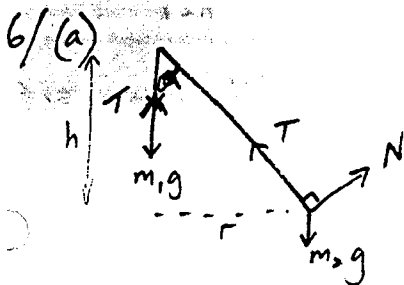
$$= 2\sqrt{3} - \int_0^{\frac{\pi}{3}} \sec \theta d\theta - \int_0^{\frac{\pi}{3}} \sec \theta \tan^2 \theta d\theta$$

$$2 \int_0^{\frac{\pi}{3}} \sec \theta \tan^2 \theta d\theta = 2\sqrt{3} - \left[\log_e(\sec \theta + \tan \theta) \right]_0^{\frac{\pi}{3}}$$

$$\int_0^{\frac{\pi}{3}} \sec \theta \tan^2 \theta d\theta = \sqrt{3} - \frac{1}{2} \log_e \left(\frac{2+\sqrt{3}}{1+0} \right)$$

$$\therefore V = 36 \left(\sqrt{3} - \frac{1}{2} \log_e (2+\sqrt{3}) \right)$$

$$= 36\sqrt{3} - 18 \log_e (2+\sqrt{3})$$



At P, $T = m_1 g$ ——— (1)

At P₂ Resolve Horizontally

$$T \sin \alpha - N \cos \alpha = m_2 r \omega^2 \text{ ——— (2)}$$

Resolve Vertically

$$T \cos \alpha + N \sin \alpha = m_2 g \text{ ——— (3)}$$

Note $\tan \alpha = \frac{r}{h} \therefore r = h \tan \alpha$ ——— (4)

$$\times \sin \alpha + (3) \times \cos \alpha$$

$$(\sin^2 \alpha + \cos^2 \alpha) = m_2 r \omega^2 \sin \alpha + m_2 g \cos \alpha, e^{kt} = \frac{g + kv}{g + kv}$$

sub in (1), (4)

$$m_1 g = m_2 h \tan \alpha \omega^2 \sin \alpha + m_2 g \cos \alpha$$

$$m_1 g \cos \alpha = m_2 h \omega^2 \sin^2 \alpha + m_2 g \cos^2 \alpha$$

$$= m_2 (h \omega^2 \sin^2 \alpha + g \cos^2 \alpha)$$

$$(ii) (2) \times \cos \alpha - (3) \times \sin \alpha$$

$$-N(\cos^2 \alpha + \sin^2 \alpha) = m_2 h \tan \alpha \omega^2 \cos \alpha - m_2 g$$

$$\therefore N = m_2 g \sin \alpha - m_2 h \omega^2 \sin \alpha$$

$$= m_2 \sin \alpha (g - h \omega^2)$$

For N to exist,

$$m_2 \sin \alpha (g - h \omega^2) > 0$$

$$\therefore g - h \omega^2 > 0$$

$$g > h \omega^2$$

(b) (i)

$$\frac{R}{m} \propto v \therefore \frac{R}{m} = kv$$

$$\therefore m \ddot{x} = -mg - mkv$$

$$\ddot{x} = -g - kv$$

$$\frac{dv}{dt} = \frac{-1}{g + kv}$$

$$t = - \int_0^v \frac{1}{g + kv} dv$$

$$= -\frac{1}{k} \left[\log_e (g + kv) \right]_0^v$$

$$= \frac{1}{k} \log_e \left(\frac{g + kv}{g} \right)$$

Make v the subject.

$$g + kv = e^{-kt}(g + kv)$$

$$\frac{dx}{dt} = v = \frac{1}{k}(e^{-kt}(g + kv) - g)$$

$$x = \frac{1}{k} \left(-\frac{1}{k} e^{-kt}(g + kv) - gt \right) + c$$

$$x = 0, t = 0 \therefore c = \frac{1}{k^2}(g + kv)$$

$$\therefore x = -\frac{1}{k^2} e^{-kt}(g + kv) - \frac{gt}{k} + \frac{1}{k^2}(g + kv)$$

$$= -\frac{gt}{k} + \frac{1}{k^2}(g + kv)(1 - e^{-kt})$$

ii) Particles meet when

$$x = h - \frac{gt}{k} - \frac{ge^{-kt}}{k^2} + \frac{g}{k^2}$$

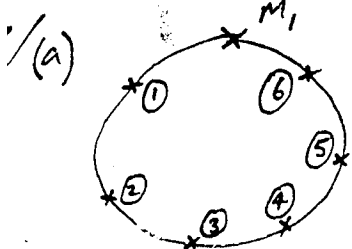
$$h - \frac{gt}{k} - \frac{ge^{-kt}}{k^2} + \frac{g}{k^2} = -\frac{gt}{k} + \frac{1}{k^2}(g + kv)(1 - e^{-kt})$$

$$h = \frac{v}{k}(1 - e^{-kt})$$

$$e^{-kt} = \frac{v - kh}{v}$$

$$-kt = \ln\left(\frac{v - kh}{v}\right)$$

$$t = \frac{1}{k} \ln\left(\frac{v}{v - kh}\right)$$



No. of ways

of seating 2 men (6)

$$\text{of table} = {}^6C_2 = 15 \quad (i) (\sqrt{a} - \sqrt{b})^2 \geq 0$$

No. of ways where they don't sit together: $[2, 4], [2, 5], [3, 5]$

= 3

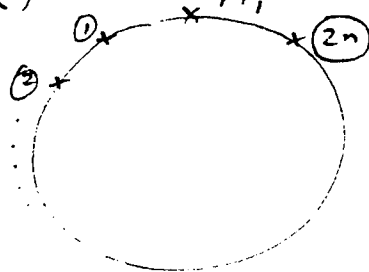
$$\therefore \text{probability} = \frac{3}{15} = \frac{1}{5}$$

$$\text{or } n(s) = {}^6P_2 = 30$$

$$n(E) = 6 \quad \left(\begin{matrix} (2, 4) & (2, 5) & (3, 5) \\ (4, 2) & (5, 2) & (5, 3) \end{matrix} \right)$$

$$\therefore P(E) = \frac{1}{5}$$

(ii)



The remaining $(n-1)$ men can sit at $2, 4, \dots, 2(n-1)$ or $3, 5, \dots, 2n-1$, or etc.

i.e. A gap (ie two women) can be n places.

No. of ways of seating $(n-1)$ men = ${}^{2n}P_{n-1}$

No. of ways which men sit apart = $n(n-1)! = n!$

$$\therefore \text{Probability} = \frac{n!}{(2n)!} \cdot \frac{(2n - (n-1))!}{1} = \frac{n!(n+1)!}{(2n)!}$$

$$\therefore a - 2\sqrt{a}\sqrt{b} + b \geq 0$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

b) cont.

ii) from (i)

$$\frac{\frac{a+b}{2} + \frac{c+d}{2}}{2} \geq \sqrt{\left(\frac{a+b}{2}\right)\left(\frac{c+d}{2}\right)}$$

$$\frac{a+b+c+d}{4} \geq \sqrt{\frac{ac+ad+bc+bd}{4}}$$

$$(a+b+c+d)^2 \geq 4(ac+ad+bc+bd) \text{--- (1)}$$

ii) Similarly

$$\frac{\frac{a+c}{2} + \frac{b+d}{2}}{2} \geq \sqrt{\left(\frac{a+c}{2}\right)\left(\frac{b+d}{2}\right)}$$

$$(a+b+c+d)^2 \geq 4(ab+ad+bc+cd) \text{--- (2)}$$

$$\frac{\frac{a+d}{2} + \frac{b+c}{2}}{2} \geq \sqrt{\left(\frac{a+d}{2}\right)\left(\frac{b+c}{2}\right)}$$

$$(a+b+c+d)^2 \geq 4(ab+ac+bd+cd) \text{--- (3)}$$

i) + (2) + (3)

$$3(a+b+c+d)^2 \geq 4(2ab+2ad+2bc+2cd+2bd+2ac)$$

$$\therefore (a+b+c+d)^2 \geq \frac{8}{3}(ab+ad+bc+cd+bd+ac)$$

c) Step 1 - prove true for $n=1$

$$\begin{aligned} \text{LHS} &= \frac{a_1 - \sqrt{2}}{a_1 + \sqrt{2}} \\ &= \frac{1 - \sqrt{2}}{1 + \sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}}\right)^{2^{1-1}} \\ &= \text{RHS} \end{aligned}$$

Step 2 Assume result true for $n=k$, k a positive integer

$$\text{i.e. } \frac{a_k - \sqrt{2}}{a_k + \sqrt{2}} = \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}}\right)^{2^{k-1}}$$

Step 3 Prove result true for $n=k+1$

$$\text{i.e. Prove } \frac{a_{k+1} - \sqrt{2}}{a_{k+1} + \sqrt{2}} = \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}}\right)^{2^{k+1}}$$

$$\text{LHS} = \frac{\frac{1}{2}\left(a_k + \frac{2}{a_k}\right) - \sqrt{2}}{\frac{1}{2}\left(a_k + \frac{2}{a_k}\right) + \sqrt{2}}$$

$$= \frac{(a_k)^2 + 2 - 2\sqrt{2}a_k}{(a_k)^2 + 2 + 2\sqrt{2}a_k}$$

$$= \left(\frac{a_k - \sqrt{2}}{a_k + \sqrt{2}}\right)^2$$

$$= \left(\left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}}\right)^{2^{k-1}}\right)^2 \quad \text{from assumpt}$$

$$= \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}}\right)^{2^k}$$

$$= \text{RHS.}$$

Step 4

The result is true for $n=1$ and, if true for $n=k$, it is true for $n=k+1$.

$\therefore$ It is true for $n=1, 2, 3, \dots$

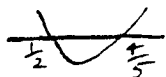
$$8/a) \quad 3\sqrt{x(1-x)} < |x-2|$$

Note: $x(1-x) \geq 0$ 
 $\therefore 0 \leq x \leq 1$

$$9x(1-x) < (x-2)^2$$

$$x^2 - 4x + 4 - 9x + 9x^2 > 0$$

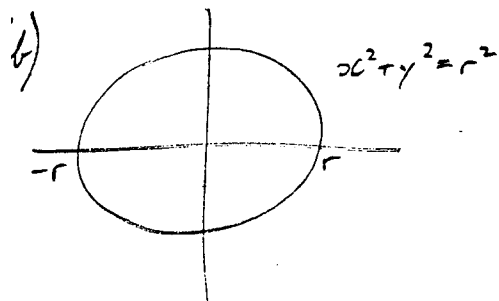
$$10x^2 - 13x + 4 > 0$$

$$(5x-4)(2x-1) > 0$$


$$\therefore x > \frac{4}{5} \text{ or } x < \frac{1}{2}$$

but $0 \leq x \leq 1$

$\therefore$ solution: $0 \leq x < \frac{1}{2}$ or $\frac{4}{5} < x \leq 1$



$$x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

length of semi-circle

$$= \int_{-r}^r \left(1 + \left(-\frac{x}{y}\right)^2\right)^{\frac{1}{2}} dx$$

$$= \int_{-r}^r \sqrt{1 + \frac{x^2}{r^2 - x^2}} dx$$

$$= 2 \int_0^r \sqrt{\frac{r^2 - x^2 + x^2}{r^2 - x^2}} dx$$

$$= 2r \int_0^r \frac{dx}{\sqrt{r^2 - x^2}}$$

$$= 2r \left[\sin^{-1} \frac{x}{r} \right]_0^r$$

$$= 2r \cdot \frac{\pi}{2}$$

$$= \pi r$$

$\therefore$ Circumference of circle $= 2\pi r$

(c) $1 - t + t^2 - t^3 + \dots + t^{2n}$

G.P. $a=1$, $r=-t$, $2n+1$ terms

$$S_{2n+1} = 1 \left(\frac{(-t)^{2n+1} - 1}{-t - 1} \right)$$

$$= \frac{1}{1+t} + \frac{(-t)^{2n+1} \cdot t}{(-t)(1+t)}$$

$$= \frac{1}{1+t} + \frac{t^{2n+1}}{1+t}$$

(ii) $\int_0^x (1 - t + t^2 - t^3 + \dots + t^{2n}) dt = \int_0^x \left(\frac{1}{1+t} + \frac{t^{2n+1}}{1+t} \right) dt$

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + \frac{x^{2n+1}}{2n+1} = \ln(1+x) + \int_0^x \frac{t^{2n+1}}{1+t} dt$$

$$\therefore \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{x^{2n+1}}{2n+1} - \int_0^x \frac{t^{2n+1}}{1+t} dt$$

(iii) $\int_0^x \frac{t^{2n+1}}{1+t} dt < \int_0^x \frac{t^{2n+1}}{t} dt$

for $x > 0$

$$\lim_{n \rightarrow \infty} \int_0^x \frac{t^{2n+1}}{t} dt$$

$$= \lim_{n \rightarrow \infty} \left(\frac{x^{2n+1}}{2n+1} \right) = 0 \text{ for } n < x$$

8(L') cont.

$$\therefore \lim_{n \rightarrow \infty} \int_0^x \frac{t^{2n+1}}{1+t} dt = 0$$

(iv) Let $x = 1$

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n+1}$$

for $n = 0, 1, 2, \dots$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$